Example 1 - Problem Statement:

Given a system of two particles:

- **Particle A_1:**
  - Mass (m_1): $1\ \mathrm{kg}$
  - Velocity ($\vec{V_1}$): $2\vec{i}\ (\mathrm{m/s})$
- **Particle A_2:**
  - Mass (m_2): $500\ \mathrm{g}$
  - Velocity ($\vec{V_2}$): $-4\vec{i}\ (\mathrm{m/s})$

Determine $\vec{V_G}$ (the velocity of the center of mass) and conclude.

Step-by-Step Solution:

Following the principles from your "Chapter 2: Linear Momentum" notes.

1. Standardize Units (SI Units): Before any calculation, we must ensure all units are in the standard SI system (kilograms, meters, seconds).

- $m_1 = 1\ \mathrm{kg}$
- $m_2 = 500\ \mathrm{g} = 0.5\ \mathrm{kg}$
- $\vec{V_1} = 2\vec{i}\ \mathrm{m/s}$
- $\vec{V_2} = -4\vec{i}\ \mathrm{m/s}$

2. Calculate the Total Linear Momentum of the System ($\vec{P_S}$): As per **Section B** of your notes, the total linear momentum of the system is the vector sum of the linear momenta of its particles.

$$\vec{P_S} = \vec{P_1} + \vec{P_2}$$

$$\vec{P_S} = m_1\vec{V_1} + m_2\vec{V_2}$$

Now, we substitute the values:

$$\vec{P_S} = (1\ \mathrm{kg}) \cdot (2\vec{i}\ \mathrm{m/s}) + (0.5\ \mathrm{kg}) \cdot (-4\vec{i}\ \mathrm{m/s})$$

$$\vec{P_S} = (2\vec{i}\ \mathrm{kg} \cdot \mathrm{m/s}) - (2\vec{i}\ \mathrm{kg} \cdot \mathrm{m/s})$$

$$\vec{P_S} = \vec{0}\ \mathrm{kg} \cdot \mathrm{m/s}$$

3. Calculate the Total Mass of the System (M_S): The total mass (M_S) is the scalar sum of the individual masses.

$$M_S = m_1 + m_2$$

$$M_S = 1\ \text{kg} + 0.5\ \text{kg}$$

$$M_S = 1.5\ \text{kg}$$

4. Determine the Velocity of the Center of Mass ($\vec{V}_G$): From **Section C** of your notes, the total linear momentum of the system is also defined by the total mass and the velocity of the center of mass:

$$\vec{P}_S = M_S \cdot \vec{V}_G$$

We can rearrange this formula to solve for $\vec{V}_G$:

$$\vec{V}_G = \frac{\vec{P}_S}{M_S}$$

Substituting our calculated values:

$$\vec{V}_G = \frac{\vec{0}\ \text{kg} \cdot \text{m/s}}{1.5\ \text{kg}}$$

$$\vec{V}_G = \vec{0}\ \text{m/s}$$

Conclusion:

The velocity of the center of mass of the system ($\vec{V}_G$) is $\vec{0}\ \text{m/s}$.

This result signifies that the center of mass of the system is stationary. Although the individual particles A_1 and A_2 are moving, their momenta are equal in magnitude ($P_1 = 2\ \text{kg} \cdot \text{m/s}$ and $P_2 = 2\ \text{kg} \cdot \text{m/s}$) and in opposite directions. As a result, the total momentum of the system is zero, and the center of mass does not move.

Example 2 - Problem Statement:

Given a system of two particles moving in a 2D plane:

- **Particle A_1:**
  - Mass (m_1): $1\ \text{kg}$
  - Velocity ($\vec{V}_1$): $(2\vec{i} - 3\vec{j})\ \text{m/s}$
- **Particle A_2:**
  - Mass (m_2): $200\ \text{g}$
  - Velocity ($\vec{V}_2$): $(-3\vec{i} + 4\vec{j})\ \text{m/s}$

Determine $\vec{V}_G$ (the velocity vector of the center of mass) and V_G (the magnitude of that velocity).

Step-by-Step Solution:

This problem follows the same principles as Example 1, but now we must handle the $\vec{i}$ (x-axis) and $\vec{j}$ (y-axis) components separately.

1. Standardize Units (SI Units): First, convert all masses to kilograms.

- $m_1 = 1 \text{ kg}$
- $m_2 = 200 \text{ g} = 0.2 \text{ kg}$

2. Calculate the Total Linear Momentum of the System ($\vec{P}_S$): Using **Section B** from your notes, we sum the individual momentum vectors:

$$\vec{P}_S = \vec{P}_1 + \vec{P}_2$$

$$\vec{P}_S = m_1\vec{V}_1 + m_2\vec{V}_2$$

Substitute the values:

$$\vec{P}_S = 1 \cdot (2\vec{i} - 3\vec{j}) + 0.2 \cdot (-3\vec{i} + 4\vec{j})$$

Distribute the scalar masses:

$$\vec{P}_S = (2\vec{i} - 3\vec{j}) + (-0.6\vec{i} + 0.8\vec{j})$$

Group the $\vec{i}$ and $\vec{j}$ components:

$$\vec{P}_S = (2 - 0.6)\vec{i} + (-3 + 0.8)\vec{j}$$

$$\vec{P}_S = 1.4\vec{i} - 2.2\vec{j} \text{ (kg} \cdot \text{m/s)}$$

3. Calculate the Total Mass of the System (M_S):

$$M_S = m_1 + m_2$$

$$M_S = 1 \text{ kg} + 0.2 \text{ kg}$$

$$M_S = 1.2 \text{ kg}$$

4. Determine the Velocity Vector of the Center of Mass ($\vec{V}_G$): Using **Section C** from your notes ($\vec{P}_S = M_S \cdot \vec{V}_G$), we solve for $\vec{V}_G$:

$$\vec{V}_G = \frac{\vec{P}_S}{M_S}$$

$$\vec{V}_G = \frac{1.4\vec{i} - 2.2\vec{j}}{1.2}$$

$$\vec{V}_G = \left(\frac{1.4}{1.2}\right)\vec{i} - \left(\frac{2.2}{1.2}\right)\vec{j}$$

$$\vec{V}_G \approx 1.17\vec{i} - 1.83\vec{j} \text{ m/s}$$

5. Determine the Magnitude of the Velocity (V_G): To find the magnitude (speed) of the center of mass, we use the Pythagorean theorem on its velocity components ($V_x \approx 1.17$ and $V_y \approx -1.83$).

$$V_G = ||\vec{V}_G|| = \sqrt{(V_x)^2 + (V_y)^2}$$

$$V_G = \sqrt{(1.17)^2 + (-1.83)^2}$$

$$V_G = \sqrt{1.3689 + 3.3489}$$

$$V_G = \sqrt{4.7178}$$

$$V_G \approx 2.17 \text{ m/s}$$

Final Answer:

- The velocity vector of the center of mass is: $\vec{V}_G \approx 1.17\vec{i} - 1.83\vec{j} \text{ m/s}$
- The magnitude of the velocity (speed) of the center of mass is: $V_G \approx 2.17 \text{ m/s}$

Application E1 - Problem Statement:

A car of mass M moves on a rough horizontal road at a velocity $\vec{V}_0$. At $t_0 = 0$, the driver brakes, and the car stops at time t_1. The frictional forces are represented by constant forces $\vec{f}$ and $\vec{f'}$, both parallel to the road and opposite to the movement.

a. Express t_1 in terms of f, f', M, and V_0. **b.** Express the traveled distance D in terms of M, V_0, f, and f'.

Part a: Finding the stopping time (t_1)

For this part, we will use the **Theorem of Linear Momentum** (also known as the Impulse-Momentum Theorem), as it relates force and time.

1. Define the System and States:

- **System:** The car of mass M.
- **Initial State (at $t_0 = 0$):**
  - Initial Velocity: $\vec{V}_0$
  - Initial Momentum: $\vec{P}_0 = M\vec{V}_0$
- **Final State (at t_1):**
  - Final Velocity: $\vec{V}_f = \vec{0}$ (the car stops)
  - Final Momentum: $\vec{P}_f = \vec{0}$
- **Time Interval:** $\Delta t = t_1 - t_0 = t_1$

2. Analyze the External Forces (ΣF_{ext}):

- Vertical Forces:
  - Weight ($\vec{W}$ or $M\vec{g}$): Acts downwards.
  - Normal Force ($\vec{N}$): Acts upwards from the road.
  - These forces cancel each other out on the horizontal road: $\vec{W} + \vec{N} = \vec{0}$.
- Horizontal Forces (Opposing Motion):
  - Friction $\vec{f}$
  - Braking force $\vec{f'}$
  - The net external force is $\Sigma\vec{F}_{ext} = \vec{W} + \vec{N} + \vec{f} + \vec{f'} = \vec{f} + \vec{f'}$.

3. Apply the Linear Momentum Theorem: From your notes (Section D, case b), for a non-isolated system with a constant external force:

$$\Delta\vec{P} = \Sigma\vec{F}_{ext} \cdot \Delta t$$

$$\vec{P}_f - \vec{P}_0 = (\vec{f} + \vec{f'}) \cdot t_1$$

4. Solve using Projections: Let's define the positive x-axis ($\vec{i}$) as the direction of the initial velocity $\vec{V}_0$.

- $\vec{V}_0 = V_0\vec{i} \implies \vec{P}_0 = MV_0\vec{i}$
- $\vec{P}_f = \vec{0}$
- Since $\vec{f}$ and $\vec{f'}$ oppose the motion, they point in the negative x-direction:
  - $\vec{f} = -f\vec{i}$
  - $\vec{f'} = -f'\vec{i}$

Now, substitute these vectors into the equation:

$$\vec{0} - (MV_0\vec{i}) = (-f\vec{i} - f'\vec{i}) \cdot t_1$$

$$-MV_0\vec{i} = -(f + f')\vec{i} \cdot t_1$$

Looking at the scalar components:

$$-MV_0 = -(f + f') \cdot t_1$$

Solving for t_1:

$$t_1 = \frac{-MV_0}{-(f + f')}$$

$$t_1 = \frac{MV_0}{f + f'}$$

Part b: Finding the stopping distance (D)

For this part, which relates force and distance, we use the **non-conservation of mechanical energy equation.** The work done by the non-conservative friction forces equals the change in the system's mechanical energy.

1. Define the States for Energy:

- **Initial State (at $x = 0$):**
  - Velocity: V_0
  - Kinetic Energy: $KE_0 = \frac{1}{2}MV_0^2$
  - Potential Energy: $GPE_0 = 0$ (we set the horizontal road as the reference level).
  - Initial Mechanical Energy: $ME_0 = \frac{1}{2}MV_0^2$
- **Final State (at $x = D$):**
  - Velocity: $V_f = 0$
  - Kinetic Energy: $KE_f = 0$
  - Potential Energy: $GPE_f = 0$
  - Final Mechanical Energy: $ME_f = 0$

2. Calculate the Work Done by Non-Conservative Forces (W_{nc}): The non-conservative forces are the two friction forces, $\vec{f}$ and $\vec{f'}$. They act over a displacement $\vec{D}$.

- Work is $W = \vec{F} \cdot \vec{D} = F \cdot D \cdot \cos(\theta)$.
- The displacement $\vec{D}$ is in the positive x-direction, while $\vec{f}$ and $\vec{f'}$ are in the negative x-direction. The angle θ is $180°$.
- $W_f = f \cdot D \cdot \cos(180°) = -fD$
- $W_{f'} = f' \cdot D \cdot \cos(180°) = -f'D$
- Total Non-Conservative Work: $W_{nc} = W_f + W_{f'} = -fD - f'D = -(f + f')D$

3. Apply the Work-Energy Theorem:

$$\Delta ME = W_{nc}$$

$$ME_f - ME_0 = W_{nc}$$

$$0 - \left(\frac{1}{2}MV_0^2\right) = -(f + f')D$$

$$-\frac{1}{2}MV_0^2 = -(f + f')D$$

Solving for D:

$$D = \frac{-\frac{1}{2}MV_0^2}{-(f + f')}$$

$$D = \frac{MV_0^2}{2(f + f')}$$

Application E2 - Problem Statement:

A ball of mass $m = 500$ g is thrown up an inclined plane of $\alpha = 0.1$ rad with an initial speed $V_A = 3$ m/s. The ball reaches point B with a speed $V_B = 1$ m/s after a time $\Delta t = 1.2$ s. The reference level for Gravitational Potential Energy (GPE) is at A. (We will use $g = 10$ m/s^2 as in your other examples).

a. Applying the general form of Newton's second law, determine the intensity of the frictional force (f). b. Determine the distance D (from A to B) traveled by the ball. c. Calculate the amount of heat produced. Convert the answer to calories.

Part a: Finding the Frictional Force (f)

We will use the **Theorem of Linear Momentum** (Impulse-Momentum) as requested, which relates force and time.

1. Define System, States, and Units:

- System: The ball.
- Mass (m): $500 \text{ g} = 0.5 \text{ kg}$
- Time Interval (Δt): 1.2 s
- Coordinate System: Let's define the positive x-axis ($\vec{i}$) as pointing **up** the incline, from A to B.
- Initial State (at A): $\vec{V}_A = 3\vec{i} \text{ m/s} \implies \vec{P}_A = m\vec{V}_A = (0.5)(3\vec{i}) = 1.5\vec{i} \text{ kg} \cdot \text{m/s}$
- Final State (at B): $\vec{V}_B = 1\vec{i} \text{ m/s} \implies \vec{P}_B = m\vec{V}_B = (0.5)(1\vec{i}) = 0.5\vec{i} \text{ kg} \cdot \text{m/s}$

2. Analyze the External Forces ($\Sigma\vec{F}_{ext}$): We only need to consider the forces parallel to the incline (along our x-axis).

- Gravitational Force (Weight) Component ($\vec{W}_x$): Acts down the incline. Its magnitude is $mg\sin(\alpha)$, so $\vec{W}_x = -mg\sin(\alpha)\vec{i}$.
- Frictional Force ($\vec{f}$): Opposes the motion (which is up the incline), so it also acts down the incline. $\vec{f} = -f\vec{i}$.
- Net Force (parallel to incline): $\Sigma\vec{F}_{ext,x} = (-mg\sin(\alpha) - f)\vec{i}$

3. Apply the Linear Momentum Theorem:

$$\Delta\vec{P} = \Sigma\vec{F}_{ext} \cdot \Delta t$$

We'll look at the x-components:

$$\vec{P}_B - \vec{P}_A = (\Sigma\vec{F}_{ext,x}) \cdot \Delta t$$

$$(0.5\vec{i}) - (1.5\vec{i}) = (-mg\sin(\alpha) - f)\vec{i} \cdot (1.2)$$

Now, looking at the scalar components:

$$-1.0 = (-mg\sin(\alpha) - f) \cdot 1.2$$

4. Solve for f:

$$-1.0 = (-(0.5)(10)\sin(0.1) - f) \cdot 1.2$$

Since $\alpha = 0.1\ \text{rad}$ *is a small angle,* $\sin(0.1) \approx 0.1$ *(or* 0.0998 *to be more precise). We'll use* 0.0998*.*

$$-1.0 = (-5 \cdot 0.0998 - f) \cdot 1.2$$

$$-1.0 = (-0.499 - f) \cdot 1.2$$

$$\frac{-1.0}{1.2} = -0.499 - f$$

$$-0.833 = -0.499 - f$$

$$f = 0.833 - 0.499$$

$$f \approx 0.334\ \text{N}$$

Part b: Finding the Distance (D)

We will use the non-conservation of ME since it relates force and distance.

$$\Delta ME = W_{nc} \implies ME_B - ME_A = W_f$$

1. Define Energy States:

- Non-Conservative Work (W_{nc}): Work done by friction $\vec{f}$. Since friction opposes the displacement $\vec{D}$, $W_f = -f \cdot D$.
- State A (Initial):
  - $KE_A = \frac{1}{2}mV_A^2 = \frac{1}{2}(0.5)(3)^2 = 2.25\ \text{J}$
  - $GPE_A = 0$ (at reference level).
  - $ME_A = 2.25\ \text{J}$
- State B (Final):
  - $KE_B = \frac{1}{2}mV_B^2 = \frac{1}{2}(0.5)(1)^2 = 0.25\ \text{J}$
  - $GPE_B = mgh_B$. The height h_B is related to D by $h_B = D\sin(\alpha)$.
  - $GPE_B = mgD\sin(\alpha) = (0.5)(10)D\sin(0.1) \approx 0.499D$.
  - $ME_B = 0.25 + 0.499D$

2. Apply the Theorem and Solve for D:

$$ME_B - ME_A = W_f$$

$$(0.25 + 0.499D) - (2.25) = -f \cdot D$$

$$0.25 + 0.499D - 2.25 = -0.334D$$

$$-2.0 + 0.499D = -0.334D$$

$$-2.0 = -0.334D - 0.499D$$

$$-2.0 = -0.833D$$

$$D = \frac{-2.0}{-0.833}$$

$$D \approx 2.4 \text{ m}$$

Part c: Finding the Heat Produced (Q)

The heat produced (Q) by friction is the energy dissipated, which is equal to the negative of the work done by friction (W_f).

1. Calculate Heat in Joules (J):

$$Q = -W_f$$

$$Q = -(-f \cdot D) = f \cdot D$$

$$Q = (0.334 \text{ N}) \cdot (2.4 \text{ m})$$

$$Q \approx 0.802 \text{ J}$$

2. Convert to Calories (cal): Given the conversion $1 \text{ cal} \approx 4.18 \text{ J}$.

$$Q \text{ (cal)} = \frac{Q \text{ (J)}}{4.18 \text{ J/cal}}$$

$$Q \text{ (cal)} = \frac{0.802}{4.18}$$

$$Q \approx 0.192 \text{ cal}$$

Application E3 - Problem Statement:

A projectile of mass $m = 100$ g is thrown from a point O with a speed $V_0 = 10$ m/s at an angle $\alpha = 30°$ *with the horizontal*, as shown. Point O is at a height $h = 1$ m above the ground. Friction is negligible. The GPE=0 reference is the ground. The y-axis is positive *downwards*. (We will use $g = 10 \text{ m/s}^2$).

a. Determine the speed of the projectile at point D just before reaching the ground. **b.** Determine the P_x and P_y components of the momentum vector at time t. **c.** Deduce the expressions of V_x and V_y at the instant t. **d.** Determine the time equations of motion $x(t)$ and $y(t)$. **e.** At what moment does the projectile reach the ground in D?

Initial Setup & Units:

- Mass (m): $100 \text{ g} = 0.1 \text{ kg}$
- Initial Velocity ($\vec{V_0}$): The vector $\vec{V_0}$ points down and right.
  - $V_{0x} = V_0 \cos(\alpha) = 10 \cdot \cos(30°) \approx 8.66 \text{ m/s}$
  - $V_{0y} = V_0 \sin(\alpha) = 10 \cdot \sin(30°) = 5 \text{ m/s}$ (positive, as it's in the +y direction)
  - $\vec{V_0} = (8.66\vec{i} + 5\vec{j}) \text{ m/s}$
- Initial Position: At $t = 0$, the projectile is at the origin O, so $x(0) = 0$ and $y(0) = 0$.
- Position of Ground (D): The ground is $h = 1\text{m}$ *below* the origin O. In our coordinate system, this is at $y = 1$ m.

a. Speed at Ground (V_D)

Since friction is negligible, mechanical energy is conserved. We'll compare the energy at O (initial) and D (final).

- Energy at O (ME_O):
  - $KE_O = \frac{1}{2}mV_0^2 = \frac{1}{2}(0.1)(10)^2 = 5$ J
  - $GPE_O = mgh_O$. Point O is 1 m *above* the GPE=0 level.
  - $GPE_O = (0.1)(10)(1) = 1$ J
  - $ME_O = KE_O + GPE_O = 5 + 1 = 6$ J
- Energy at D (ME_D):
  - $KE_D = \frac{1}{2}mV_D^2 = \frac{1}{2}(0.1)V_D^2 = 0.05V_D^2$
  - $GPE_D = 0$ (at the reference level).
  - $ME_D = 0.05V_D^2$
- Conservation of Mechanical Energy ($ME_O = ME_D$):
  - $6 = 0.05V_D^2$
  - $V_D^2 = \frac{6}{0.05} = 120$
  - $V_D = \sqrt{120} \approx 10.95 \text{ m/s}$

b. Momentum Components (P_x, P_y) at time t

We use the Theorem of Linear Momentum: $\Delta\vec{P} = \Sigma\vec{F}_{ext} \cdot \Delta t$. The only external force is gravity: $\Sigma\vec{F}_{ext} = m\vec{g}$. Since the +y axis is downwards, $\vec{g} = g\vec{j} = 10\vec{j}$.

- $P_t - P_0 = (m\vec{g}) \cdot t$
- $\vec{P_t} = \vec{P_0} + (m\vec{g})t$

First, find the initial momentum vector $\vec{P_0}$:

- $\vec{P_0} = m\vec{V_0} = 0.1 \cdot (8.66\vec{i} + 5\vec{j}) = 0.866\vec{i} + 0.5\vec{j}$

Now, substitute into the theorem:

- $\vec{P_t} = (P_x\vec{i} + P_y\vec{j}) = (0.866\vec{i} + 0.5\vec{j}) + (0.1 \cdot 10\vec{j})t$
- $\vec{P_t} = (0.866\vec{i} + 0.5\vec{j}) + (1\vec{j})t$
- $\vec{P_t} = 0.866\vec{i} + (0.5 + t)\vec{j}$

By comparing components, we get:

- $P_x = 0.866 \text{ kg} \cdot \text{m/s}$ (This is constant, as there is no horizontal force)
- $P_y = 0.5 + t \text{ kg} \cdot \text{m/s}$

c. Velocity Components (V_x, V_y) at time t

We can find the velocity components directly from the momentum components, since $\vec{P} = m\vec{V} \implies \vec{V} = \frac{\vec{P}}{m}$.

- $V_x = \frac{P_x}{m} = \frac{0.866}{0.1} = 8.66 \text{ m/s}$ (Constant, as expected)
- $V_y = \frac{P_y}{m} = \frac{0.5+t}{0.1} = 5 + 10t \text{ m/s}$

d. Equations of Motion ($x(t), y(t)$)

We find the position equations by taking the anti-derivative (integral) of the velocity components.

- **x-position:**
  - $V_x = \frac{dx}{dt} = 8.66$
  - $x(t) = \int 8.66 \, dt = 8.66t + C_1$
  - At $t = 0$, $x = 0$ (at origin O). So $C_1 = 0$.
  - $x(t) = 8.66t$
- **y-position:**
  - $V_y = \frac{dy}{dt} = 5 + 10t$
  - $y(t) = \int (5 + 10t) \, dt = 5t + 10\left(\frac{t^2}{2}\right) + C_2 = 5t + 5t^2 + C_2$
  - At $t = 0$, $y = 0$ (at origin O). So $C_2 = 0$.
  - $y(t) = 5t^2 + 5t$

e. Time to Reach Ground (t_D)

The projectile reaches the ground at point D. From our setup, we know the ground is at the y-coordinate $y = 1 \text{ m}$. We just need to find the time t_D when $y(t_D) = 1$.

- $y(t_D) = 5t_D^2 + 5t_D$
- $1 = 5t_D^2 + 5t_D$
- $5t_D^2 + 5t_D - 1 = 0$

This is a quadratic equation. We use the formula $t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

- $t_D = \frac{-5 \pm \sqrt{5^2 - 4(5)(-1)}}{2(5)}$
- $t_D = \frac{-5 \pm \sqrt{25 + 20}}{10}$